Trading System Data Requirements

1. Overview

This document defines the data the trading bot needs for Polymarket's 5-minute crypto Up/Down markets (for example `btc-updown-5m-<t>`): what each item looks like, where it comes from, and why it is needed.

- **Data covered:** order book (live buy and sell offers), market information (which market is open now), asset price (the BTC price), order fills (our completed trades) and market result (who won).
- **How a market resolves:** each market lasts 5 minutes. If the price at the end is the same or higher than the price at the start (the price to beat), UP wins. If it is lower, DOWN wins. A tie is UP.
- **Price source:** Polymarket decides the winner with the Chainlink BTC/USD price, not with exchange prices. So exchange prices (Binance, Coinbase, etc.) are only a close estimate.

2. Data Sources

Data	Source
Order book	CLOB REST + Market WebSocket
Market info	Gamma API
Chainlink BTC/USD (resolution price)	Polymarket price stream
Proxy price	Binance / Coinbase / OKX / Bybit
Market result	Market WebSocket `market_resolved`
Order fills (later)	User WebSocket

3. Data Details

3.1 Order Book

```
{
  "market": "0xCONDITION_ID",
  "asset_id": "TOKEN_ID",
  "bids": [{ "price": "0.47", "size": "2500" }, { "price": "0.48", "size":
"1000" }],
  "asks": [{ "price": "0.53", "size": "1500" }, { "price": "0.52", "size":
"800" }],
  "tick_size": "0.01",
  "min_order_size": "5",
  "neg_risk": false,
  "last_trade_price": "0.48",
  "timestamp": "1234567890",
  "hash": "a1b2..."
}
```

Field	Meaning
`market`	The ID of the whole market (also called the condition ID). One market has two tokens: UP and DOWN.
`asset_id`	The ID of one token (UP or DOWN). This order book belongs to this token.
`bids`	Buy offers. Each one is a price people want to buy at, with the amount available at that price.
`asks`	Sell offers. Each one is a price people want to sell at, with the amount available at that price.
`price`	Price of one share, from 0 to 1. For example, 0.48 means 48 cents, or about a 48% chance.
`size`	Number of shares available at that price.
`tick_size`	The smallest price step allowed. 0.01 means prices move in steps of 1 cent.
`min_order_size`	The smallest order allowed, in shares.

Field	Meaning
`neg_risk`	`false` for a normal market. `true` only for special multi-outcome markets.
`last_trade_price`	The price of the most recent trade on this token.
`timestamp`	The time when this order book snapshot was made (unix time).
`hash`	A short code (fingerprint) of the book. If it changes, the book has changed.

Source:

- REST: `https://clob.polymarket.com/book?token_id=TOKEN_ID`
- WebSocket: `wss://ws-subscriptions-clob.polymarket.com/ws/market`

```
{ "assets_ids": ["UP_TOKEN_ID", "DOWN_TOKEN_ID"], "type": "market",
"custom_feature_enabled": true }
```

Field	Meaning
`assets_ids`	The list of token IDs we want live updates for (UP and DOWN).
`type`	The channel type. `market` means public market data.
`custom_feature_enabled`	`true` turns on extra events: `best_bid_ask`, `new_market` and `market_resolved`.

`custom_feature_enabled` is required for `best_bid_ask`, `new_market` and `market_resolved`.

Why we need it: The order book shows the live buy and sell prices and how many shares are available at each price. We use it to see the best price now and to estimate at what price our trade would be filled.

3.2 Market Information

```
{
  "slug": "btc-updown-5m-1775241600",
  "conditionId": "0xCONDITION_ID",
  "upTokenId": "UP_TOKEN_ID",
  "downTokenId": "DOWN_TOKEN_ID",
  "windowStart": 1775241600,
  "windowEnd": 1775241900,
  "active": true
}
```

Field	Meaning
`slug`	The unique name of the market. The number at the end is the window start time.
`conditionId`	The ID of the market. It is the same as `market` in the order book.
`upTokenId`	The token ID of the UP outcome. We read its order book with this ID.
`downTokenId`	The token ID of the DOWN outcome.
`windowStart`	The start time of the 5-minute window (unix seconds). The price to beat is taken at this time.
`windowEnd`	The end time of the window (unix seconds). It is the start time + 300.
`active`	`true` if the market is active (not closed).

Source:

- `https://gamma-api.polymarket.com/events`
- `https://gamma-api.polymarket.com/markets`

Why we need it: The market information tells us which 5-minute window is open now, when it starts and ends, and the token IDs of UP and DOWN. With these IDs we know which order books to read.

3.3 Asset Price

```
{ "symbol": "BTC", "ts": 1775241600000, "chainlink": 76030.01, "binance": 76028.50, "coinbase": 76031.20 }
```

Field	Meaning
`symbol`	The asset, for example BTC.
`ts`	The time of this price record (unix milliseconds).
`chainlink`	The Chainlink BTC/USD price. This is the price that decides the winner.
`binance`	The BTC price on Binance (quoted in USDT). Only an estimate.
`coinbase`	The BTC price on Coinbase (quoted in USD). Only an estimate.

Source:

- Chainlink (new): `wss://ws-live-v2.polymarket.com/ws`
- Chainlink (legacy, deprecated): `wss://ws-live-data.polymarket.com`
- Exchanges: Binance, Coinbase, OKX and Bybit public price APIs

Why we need it: The asset price gives us the price to beat at the start of the window and the close price at the end. Chainlink decides the winner, and exchange prices give us a fast estimate.

3.4 Order Status and Fills (later)

Not needed until order placing is added. Estimate fills from the book: BUY walks the asks upward, SELL walks the bids downward. Respect tick size and min order size.

```
{ "event_type": "trade", "id": "TRADE_ID", "taker_order_id": "ORDER_ID", "asset_id": "TOKEN_ID", "side": "BUY", "size": "10", "price": "0.52", "status": "MATCHED" }
```

Field	Meaning
`event_type`	The type of message. `trade` means a trade happened.
`id`	The ID of this trade.
`taker_order_id`	The ID of the order that took the existing offers (the taker order).
`asset_id`	The token that was traded (UP or DOWN).
`side`	`BUY` or `SELL`.
`size`	Number of shares in the trade.
`price`	The price of one share in the trade.
`status`	The stage of the trade: MATCHED, MINED, CONFIRMED, RETRYING or FAILED.

Source: `wss://ws-subscriptions-clob.polymarket.com/ws/user`

Why we need it: Order fills tell us if our order was filled, how many shares we got and at what price. We need this to track our real position and balance once the bot places orders.

3.5 Market Result


```
{ "slug": "btc-updown-5m-1775241600", "priceToBeat": 76030.01, "closePrice": 76045.20, "result": "UP", "source": "computed" }
```


Field	Meaning
`slug`	The market name (same as in 3.2).
`priceToBeat`	The Chainlink price at window start. The end price must be equal or higher for UP.
`closePrice`	The Chainlink price at window end.
`result`	The winner: `UP` or `DOWN`.
`source`	Where the result came from: `computed` (our own calculation) or `polymarket` (official event).

Why we need it: The market result tells us if UP or DOWN won. We use it to score each trade as a win or a loss and to calculate the profit and loss (P&L) of the bot.

4. Open Items

- Chainlink feed needs API credentials. Decide: create the credentials, or use exchange prices instead (results marked `estimated`).
- On small price moves, exchange prices can show a different winner than Chainlink.
- Polymarket rate limits (how many requests are allowed) are not checked yet.

5. Conclusion

All data for the trading bot comes from Polymarket CLOB and Gamma plus public exchange APIs, except Polymarket's Chainlink feed, which now needs API credentials. Fill data is needed only once order placing is added.